Lab Exercise 3- Working with Docker Networking

Step 1: Understanding Docker Default Networks

Docker provides three default networks:

- bridge: The default network when a container starts.
- host: Bypasses Docker's network isolation and attaches the container directly to the host network.
- none: No networking is available for the container.

1.1. Inspect Default Networks

Check Docker's default networks using:

```
docker network ls
```

1.2. Inspect the Bridge Network

```
docker network inspect bridge
```

This command will show detailed information about the bridge network, including the connected containers and IP address ranges.

Step 2: Create and Use a Bridge Network

2.1. Create a User-Defined Bridge Network

A user-defined bridge network allows containers to communicate by name instead of IP.

```
docker network create my_bridge
```

2.2. Run Containers on the User-Defined Network

Start two containers on the newly created my_bridge network:

```
docker run -dit --name container1 --network my_bridge busybox

docker run -dit --name container2 --network my_bridge busybox
```

2.3. Test Container Communication

Execute a ping command from container1 to container2 using container names:

```
docker exec -it container1 ping container2
```

The containers should be able to communicate since they are on the same network.

Step 3: Disconnect and Remove Networks

3.1. Disconnect Containers from Networks

To disconnect container1 from my_bridge:

```
docker network disconnect my_bridge container1
```

4.2. Remove Networks

To remove the user-defined network:

```
docker network rm my_bridge
```

Step 4: Clean Up

Stop and remove all containers created during this exercise:

```
docker rm  -f container1 container2
```

```
NK@DESKTOP-M3UHPH7 MINGW64 ~
$ docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
95be618e93a5   bridge    bridge    local
c90825f6fbe4   host      host      local
7bf297bc4f50   none      null      local

NK@DESKTOP-M3UHPH7 MINGW64 ~
$ docker network inspect bridge
[
    {
        "Name": "bridge",
        "Id": "95be618e93a5844dd128f0e54eceed45da62908c2f65409847652e79eb761afb"
,
        "Created": "2026-02-24T08:58:18.1672201Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.17.0.0/16",
                    "IPRange": "",
                    "Gateway": "172.17.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.bridge.default_bridge": "true",
            "com.docker.network.bridge.enable_icc": "true",
            "com.docker.network.bridge.enable_ip_masquerade": "true",
            "com.docker.network.bridge.host_binding_ipv4": "0.0.0.0",
            "com.docker.network.bridge.name": "docker0",
            "com.docker.network.driver.mtu": "1500"
        },
        "Labels": {},
        "Containers": {},
        "Status": {
            "IPAM": {
                "Subnets": {
                    "172.17.0.0/16": {
```

```
NK@DESKTOP-M3UHPH7 MINGW64 ~
$ docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
95be618e93a5   bridge    bridge    local
c90825f6fbe4   host      host      local
7bf297bc4f50   none      null      local

NK@DESKTOP-M3UHPH7 MINGW64 ~
$ docker network create my_bridge
3851fd7e190ecd1f32a75e833c107a0fcc64f75404f7875cabc458be8e369179

NK@DESKTOP-M3UHPH7 MINGW64 ~
$ docker network ls
NETWORK ID     NAME        DRIVER    SCOPE
95be618e93a5   bridge      bridge    local
c90825f6fbe4   host        host      local
3851fd7e190e   my_bridge   bridge    local
7bf297bc4f50   none        null      local

NK@DESKTOP-M3UHPH7 MINGW64 ~
$ docker run -dit --name container1 --network my_bridge busybox
Unable to find image 'busybox:latest' locally
latest: Pulling from library/busybox
61dfb50712f5: Pulling fs layer
61dfb50712f5: Download complete
61dfb50712f5: Pull complete
96cfb76e59bd: Download complete
Digest: sha256:b3255e7dfbcd10cb367af0d409747d511aeb66dfac98cf30e97e87e4207dd76f
Status: Downloaded newer image for busybox:latest
c78dfd474aeba95b8020bad4a98f68f921fc95cabebd3f7eec2d799940b239df

NK@DESKTOP-M3UHPH7 MINGW64 ~
$ docker run -dit --name container2 --network my_bridge busybox
fc9a74543b645df864b0ae16894c104e373516edc522752f7dfc52fed29c100f
```

```
64 bytes from 172.18.0.3: seq=154 ttl=64 time=0.062 ms
64 bytes from 172.18.0.3: seq=155 ttl=64 time=0.058 ms
64 bytes from 172.18.0.3: seq=156 ttl=64 time=0.056 ms
64 bytes from 172.18.0.3: seq=157 ttl=64 time=0.060 ms
64 bytes from 172.18.0.3: seq=158 ttl=64 time=0.059 ms
64 bytes from 172.18.0.3: seq=159 ttl=64 time=0.095 ms
64 bytes from 172.18.0.3: seq=160 ttl=64 time=0.088 ms
64 bytes from 172.18.0.3: seq=161 ttl=64 time=0.064 ms
^C
--- container2 ping statistics ---
162 packets transmitted, 162 packets received, 0% packet loss
round-trip min/avg/max = 0.055/0.071/0.221 ms

NK@DESKTOP-M3UHPH7 MINGW64 ~
$ docker network disconnect my_bridge container1

NK@DESKTOP-M3UHPH7 MINGW64 ~
$ docker network rm my_bridge
Error response from daemon: error while removing network: network my_bridge has active endpoints (name:"container2" id:"2dc620d041b2")
exit status 1

NK@DESKTOP-M3UHPH7 MINGW64 ~
$ docker rm -f container1 container2
container1
container2

NK@DESKTOP-M3UHPH7 MINGW64 ~
$ docker network rm my_bridge
my_bridge

NK@DESKTOP-M3UHPH7 MINGW64 ~
```